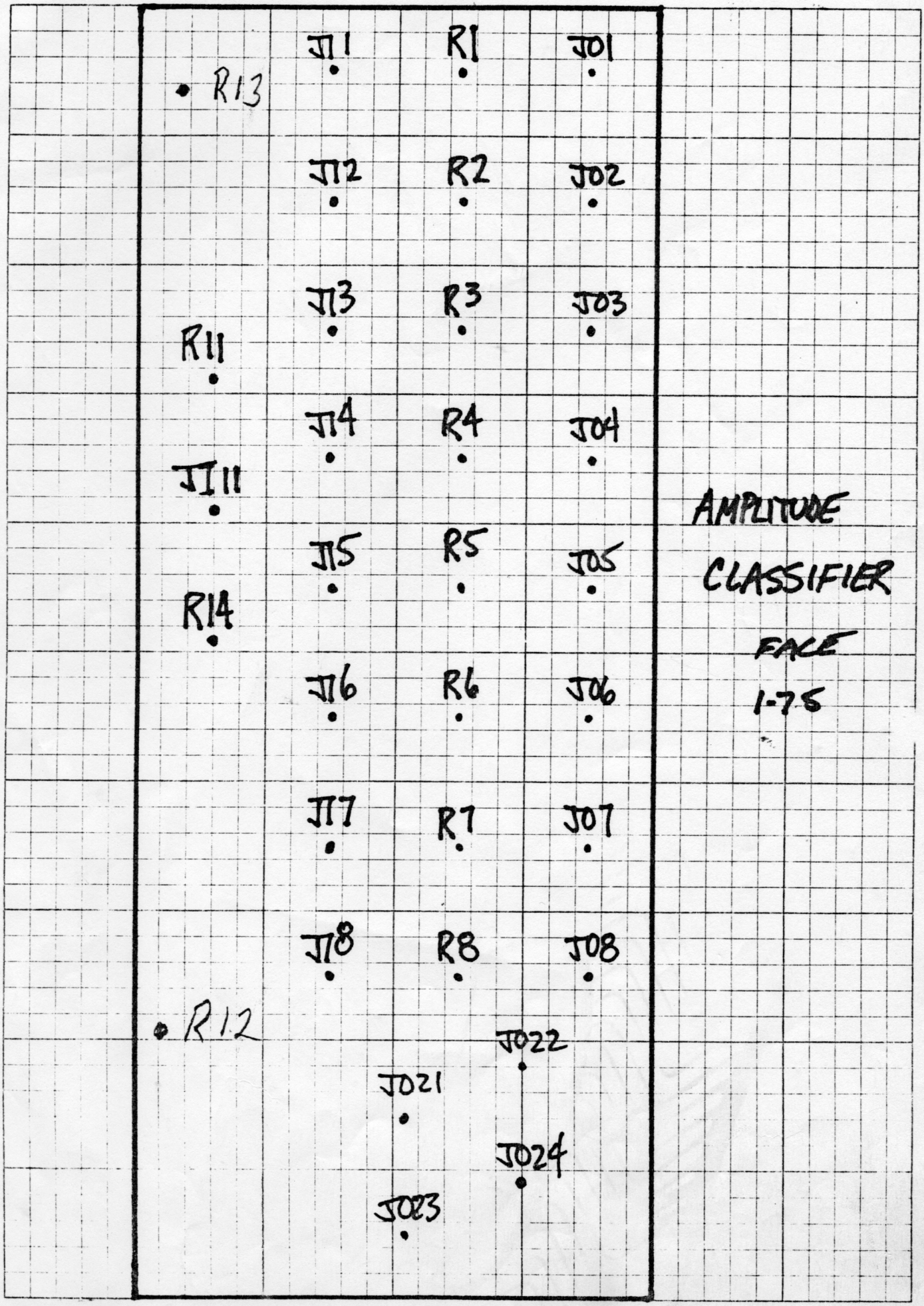
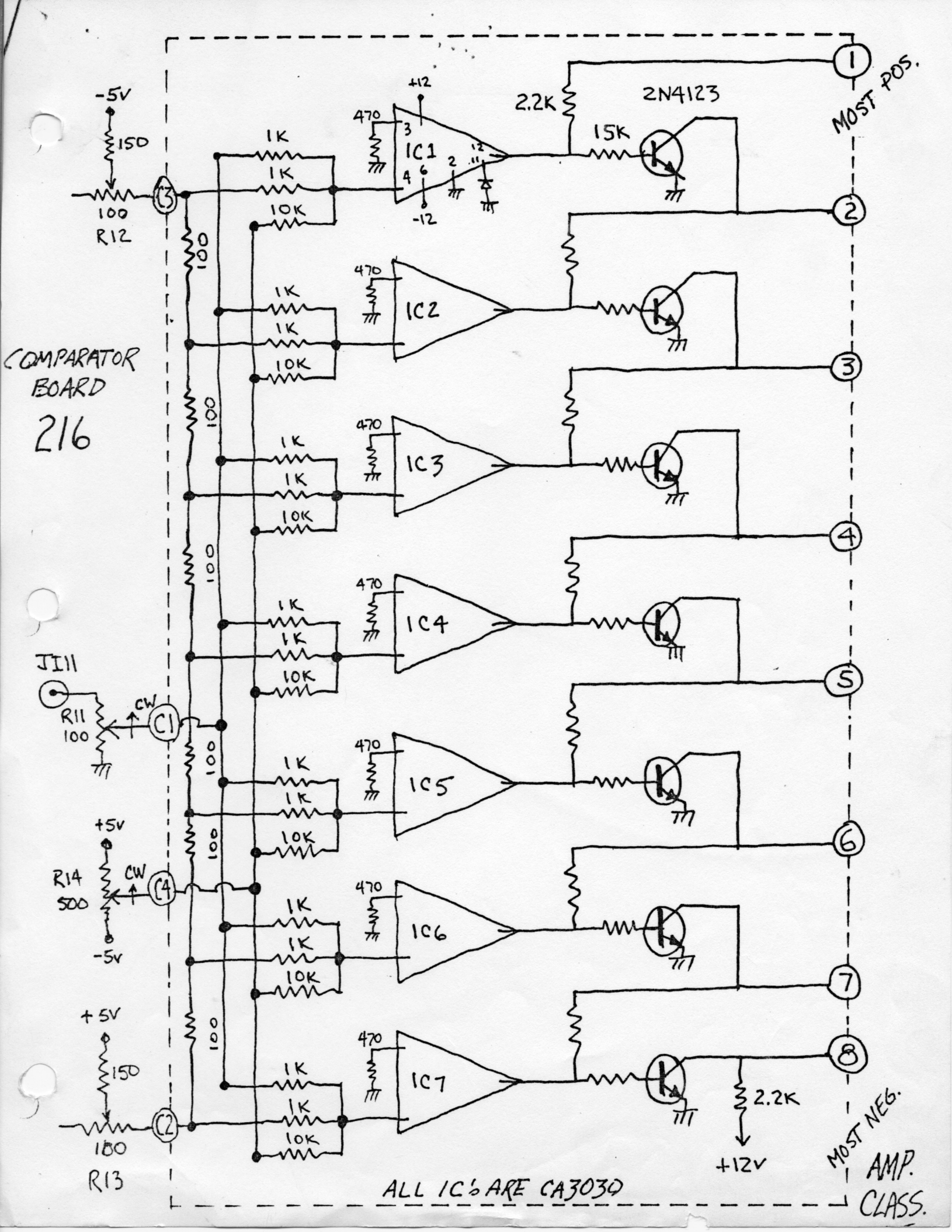
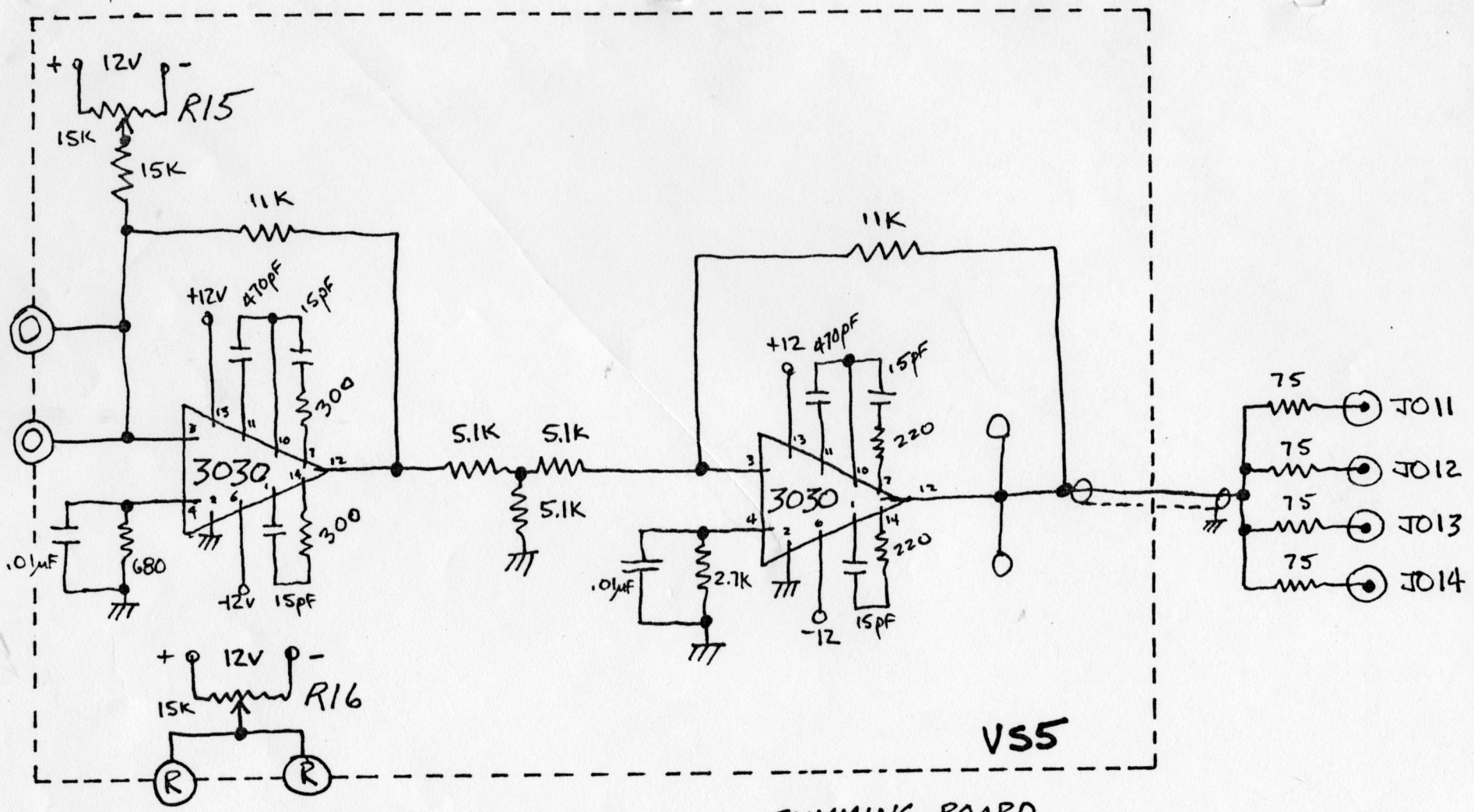

J1
R1
J01
R13
J12
R2
J02
J13
R3
J03
R11
J14
R4
J04
J11
J15
R5
J05
R14
AMPLITUDE
CLASSIFIER
FACE
1-75
J16
R6
J06
J17
R7
J07
J18
R8
J08
R12
J022
J021
J024
J023

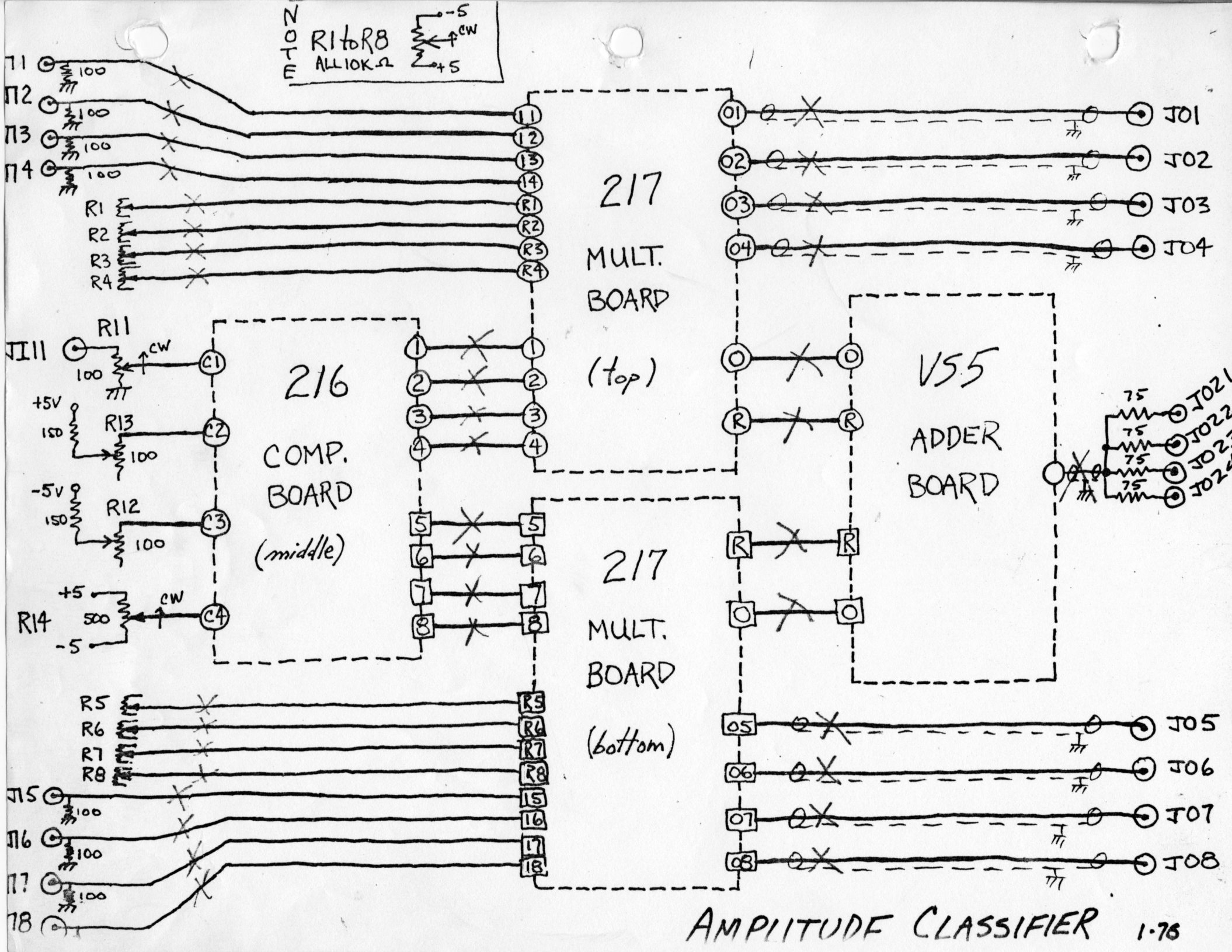

COMPARATOR BOARD
216
-5V
150
100
R12
+12
-12
470
IC1
IC2
IC3
IC4
IC5
IC6
IC7
1K
10K
100
2.2K
15K
2N4123
MOST POS.
MOST NEG.
AMP. CLASS.
+12V
J111
R11
100
CW
+5V
R14
500
-5V
+5V
150
100
R13
ALL IC's ARE CA3030

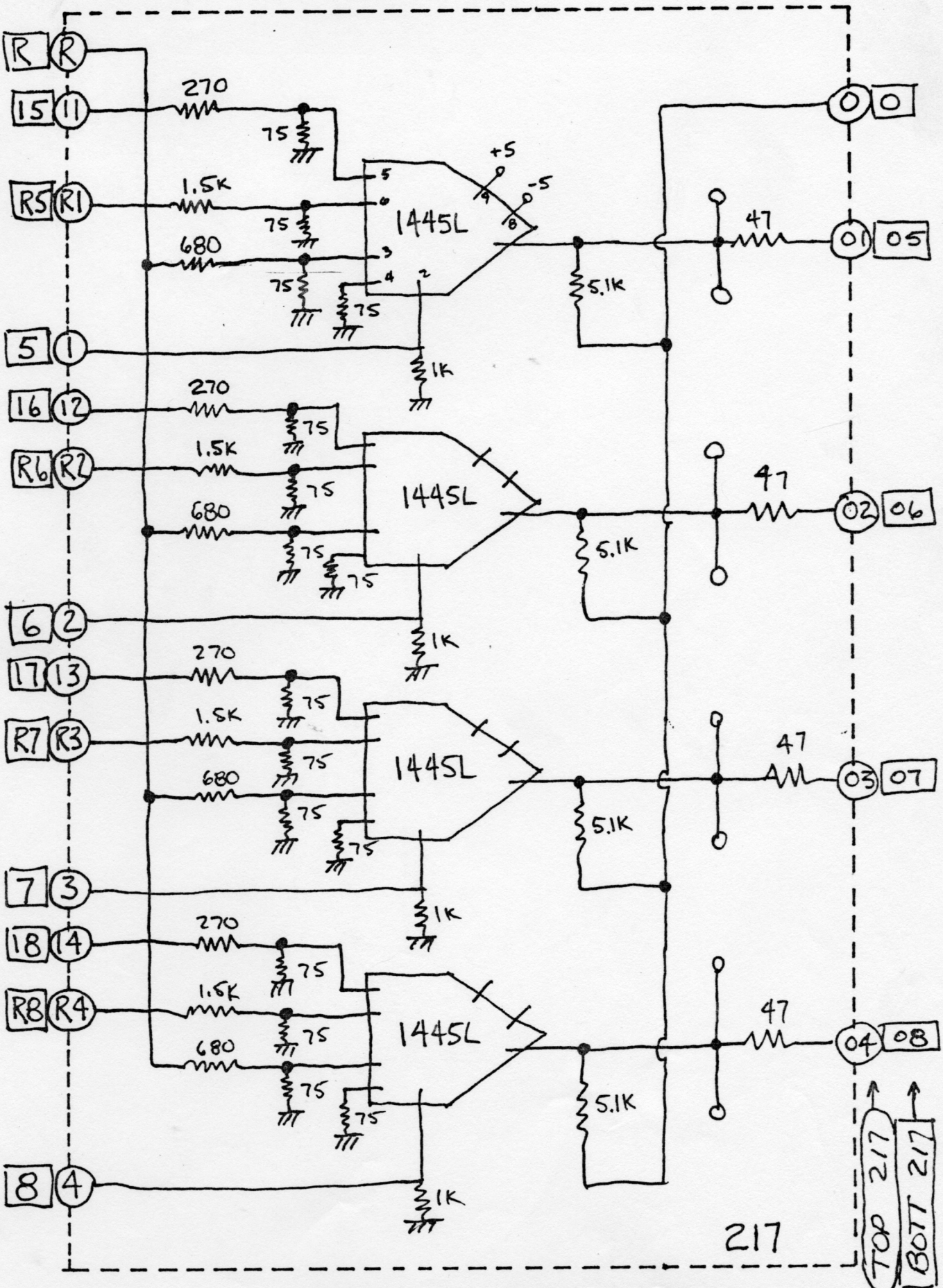


SUMMING BOARD

AMP. CLASS.

1-76


NOTE
R1 to R8
ALL 10K Ω
-5
CW
+5
J1
J2
J3
J4
100
R1
R2
R3
R4
11
12
13
14
R1
R2
R3
R4
217
MULT.
BOARD
(top)
01
02
03
04
J01
J02
J03
J04
R11
J11
CW
C1
216
COMP.
BOARD
(middle)
+5V
150
R13
C2
-5V
150
R12
C3
R14
+5
500
CW
C4
-5
1
2
3
4
5
6
7
8
O
R
V55
ADDER
BOARD
75
J21
J22
J23
J24
217
MULT.
BOARD
(bottom)
R5
R6
R7
R8
J5
J6
J7
J8
15
16
17
18
05
06
07
08
J05
J06
J07
J08
AMPLITUDE CLASSIFIER
1-76



MULTIPLIER BOARD

TOP OR BOTT.

AMP. CLASS.

1-76

OSCILLATOR

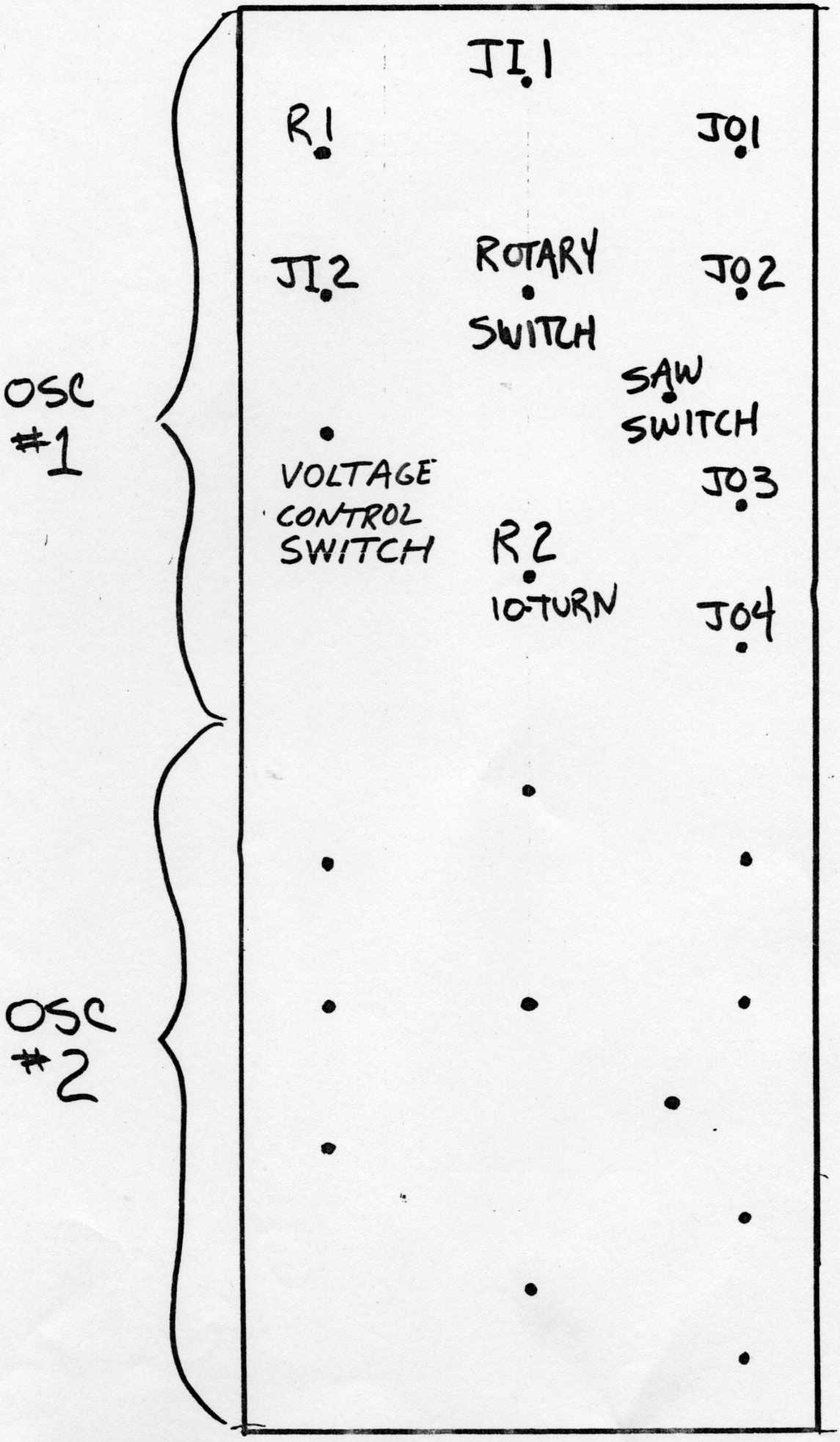
This module contains two oscillators that generates a sine wave output available at JO3, JO4 and a triangle wave output at JO1 and JO2. If the sawtooth switch is down instead of up, the triangle wave becomes a sawtooth and the sine wave becomes an "s" wave. The rotary switch sets the gross frequency range form 1/100 Hz. to ½ MHz. R2 is the continuous frequency adjustment. If the voltage control switch is up, a signal inputed to JI2 will control the frequence of the oscillator in combination with R2. With the switch down the voltage control is disabled but the oscillator is more stable. A sync. level (4 volt) signal into JI1 will trigger the oscillator to stabilize patterns.

CIRCUIT DESCRIPTION

The 8038 is a complete voltage controlled oscillator whose frequency is controlled by resistance R2 and the voltage at pin 8. The 715 is a 10x amplifier and- in combination with the zener diode produces a controll voltage at pin 8.

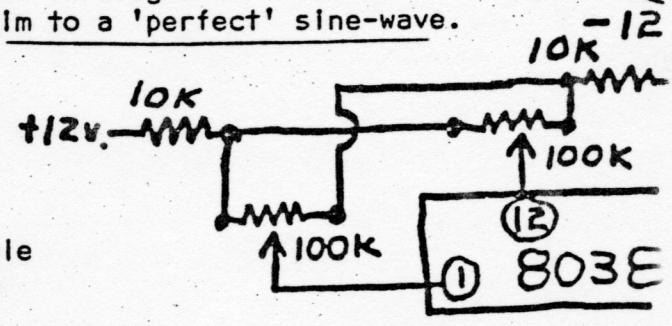
The trimmer associated with the input of the 715 should be adjusted so that the control voltage is centered within its range with 0 volts in. To do this, input a triangle wave to the voltage control input and adjust the trimmer until the voltage that makes the maximum frequency is as positive as the voltage that quenches the oscillator is negative.

The two transistors and zeners are used to trigger the oscillator. When a fast-falling signal is presented at the sync. input this turns the first transistor off which turns the second transistor on. This clamps the oscillator to the bottom of its output wave form.


JI.1
R1
JO1
JI.2
ROTARY
SWITCH
JO2
SAW
SWITCH
JO3
VOLTAGE
CONTROL
SWITCH
R2
10-TURN
JO4
OSC
#1
OSC
#2
FRONT
FACE
OSCILLATOR
9.75

1.) SINE-WAVE PURITY CONTROL:

Remove 82K resistor; and, add 100K trim-pots as shown in diagram. These 100K trim-pots correct sine-wave purity. You should be able to trim to a 'perfect' sine-wave.

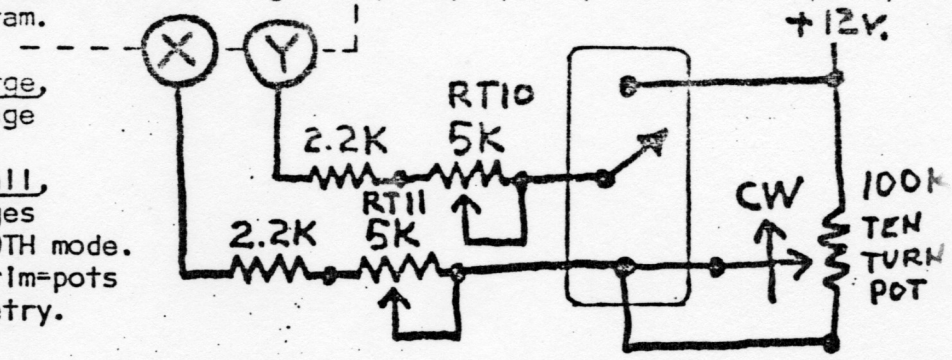


PROCEDURE-

A) Before supplying power to the module, center all trim-pots.
B) Set the oscillator at a middle freguency range, and display sine-wave on scope.
C) Tweek the trim-pots for highest amplitude possible (±1 volt) without creating any flats or peaks in the waveform; i.e. 'perfect' sine-wave.

2.) HIGH-FREQUENCY SYMETRY CONTROL:

R10 and R11 maybe replaced by a series combination of 2.2K resistor and a 5K trim-pot. This series combination (RT10 and RT11) correct high-frequency symetry and low-frequency quenching of waveform; see diagram.



If both trim-pots are too large, the high-frequency end of each range will be lower than optimum.

If both trim-pots are too small, the low-frequency end in some ranges may quench, particularly in SAWTOOTH mode.

The difference between the trim=pots determines the high-frequency symetry.

PROCEDURE-

A) All trim-pots should have been centered in above procedure, before supplying power; if, you didn't you might have blown the 8038...!
B) Turn 10-turn pot to extreme left (lowest freq.); check to make sure that no range quenc in sawtooth mode. If quenching happens in any range, tweek trim-pot to get rid of it...
C) Turn 10-turn pot to extreme right (highest-freq.); check to make sure that in a higher frequency range you still have good symetry in triangle mode. If you don't have good triar symetry, tweek trim-pot to get it...

GO BACK AND CHECK FOR SAWTOOTH QUENCHING...

D) To maximize high-frequency in ranges, decrease both trim-pots equally and go-to-step B). If oscillator quenches at low-frequencies, back up some; i.e. increase resistance, go-to step C). Stop.

NOTE:

These trim-pots will have to be outboarded on a perf-board and attached to card support fr of the module. Leave enough lead length on the trim-pots so it can be gotten out of the w for servicing the cards...!

Some 8038 integrated circuits appear to behave better than others; you may want to try various 8038's, choosing the best behaved ones...!